CONSERVED CHARGES IN LINEARLY REINFORCED GRAPH URNS

YITZCHAK SHMALO

Abstract. In the graph-based Pólya urn with linear reinforcement (WARM with $\alpha = 1$), every vector σ in the kernel of the unsigned vertex–edge incidence matrix yields an exact martingale $\sum_e \sigma_e H(N_e - 1)$, where H is the harmonic number. These charges are the flat directions of the Lyapunov function, and they determine which equilibrium the process selects. On the even cycle C_L the limit is $(1+u, 1-u, \dots)$ with $L\operatorname{artanh}(u)$ equal to the terminal value of the charge, and $\sqrt{L}\operatorname{artanh}(u)$ converges in law to a centred Gaussian as $L \to \infty$. For large even cycles this describes the law of the random limit, in the continuous-time form of the question raised by Chen and Lucas and by Lima for the discrete-time urn.

1. The model and the result

In the urn with graph-based competition of van der Hofstad, Holmes, Kuznetsov and Ruszel [4] (WARM), each edge of a graph carries a count, each vertex carries a Poisson clock, and when a vertex rings it reinforces one of its edges with probability proportional to N_e^α. For $\alpha > 1$ the counts localise; at $\alpha = 1$ the process sits at the boundary of that regime. The discrete-time urn, in which every edge adds a ball to one of its two endpoints at each step, was introduced by Benaïm, Benjamini, Chen and Lima [2]. At $\alpha = 1$ it was studied by Chen and Lucas [3], who proved convergence on regular bipartite graphs to a random point of a segment of equilibria and left the law of that point open, and by Lima [5], who completed the convergence proof for balanced bipartite graphs and asked for the distribution of the limit (his Question 1). On the infinite line, Kleptsyn conjectures that the normalised counts of the critical process converge to 1 [1, Section 5].

The random limit is fixed by conserved charges. For every σ in the kernel of the unsigned incidence matrix, $\sum_e \sigma_e H(N_e - 1)$ is an exact martingale (Theorem 1); up to a vanishing error it equals $\sum_e \sigma_e \log(N_e/t)$, which is conserved by the mean-field flow. On the even cycle C_L the kernel is one-dimensional and the limit is $(1+u_\infty, 1-u_\infty, \dots)$ with $L\operatorname{artanh}(u_\infty)$ equal to the terminal value M_∞ of the charge (Theorem 4). The quadratic variation of M is a sum of local terms, and a law of large numbers for it gives $\sqrt{L}\operatorname{artanh}(u_\infty) \Rightarrow \mathcal{N}(0, c)$, where c is the expected quadratic variation per vertex of the process on $\mathbb{Z}$ (Theorem 8). Section 6 reports simulations: the predicted variance on cycles, random limits on C_4 and K_4 whose dimension

matches the kernel, and the slow decay of the alternating envelope on long cycles that the charges predict for the line.

2. The charge martingales

Let $G = (V, E)$ be a finite graph. Each vertex carries an independent rate-one Poisson clock; when the clock of v rings, one edge $e \ni v$ is chosen with probability N_e/S_v, where $S_v = \sum_{f \ni v} N_f$, and N_e increases by one. All counts start at 1. Write $X_e(t) = N_e(t)/t$ and let B be the unsigned incidence matrix, $B_{ve} = 1$ when $v \in e$.

Theorem 1. *Let $\sigma \in \mathbb{R}^E$ satisfy $\sum_{e \ni v} \sigma_e = 0$ for every v. Then*

$$M_\sigma(t) = \sum_{e \in E} \sigma_e \, H(N_e(t) - 1), \qquad H(n) = \sum_{k=1}^{n} \frac{1}{k},$$

is a martingale with $M_\sigma(0) = 0$, jumps bounded by $\max_e |\sigma_e|$, and predictable quadratic variation

$$\langle M_\sigma \rangle_t = \int_0^t \sum_{v \in V} \frac{1}{S_v(s)} \sum_{e \ni v} \frac{\sigma_e^2}{N_e(s)} \, ds.$$

Proof. When N_e moves from n to $n+1$, $H(N_e - 1)$ increases by $1/n$. A ring at v therefore changes M_σ by σ_e/N_e with probability N_e/S_v, for each $e \ni v$. The conditional mean of the change is $S_v^{-1} \sum_{e \ni v} \sigma_e = 0$, and its conditional second moment is $S_v^{-1} \sum_{e \ni v} \sigma_e^2/N_e$. The jump rates are bounded by $|V|$ and the jumps by $\max_e |\sigma_e|$, so the compensated sums are martingales. □

The space of such σ is $\ker B$, of dimension $|E| - |V| + \beta$, with β the number of bipartite components. It is nonzero exactly when some component contains an even cycle or two odd cycles. Summing the vertex constraints gives $\sum_e \sigma_e = 0$ for $\sigma \in \ker B$, and $H(n-1) = \psi(n) + \gamma$, so

$$\sum_e \sigma_e \log X_e(t) = M_\sigma(t) + \sum_e \sigma_e \big(\log N_e(t) - \psi(N_e(t))\big), \tag{1}$$

with $\log n - \psi(n) = \frac{1}{2n} + O(n^{-2})$. The charges $\sum_e \sigma_e \log X_e$ are therefore martingales up to a term that vanishes as every count grows. In the mean-field equation $\dot{x}_e = x_e \sum_{v \in e} S_v^{-1} - x_e$ (time $\log t$) they are exactly conserved: $\frac{d}{d\tau} \sum_e \sigma_e \log x_e = \sum_v S_v^{-1} \sum_{e \ni v} \sigma_e - \sum_e \sigma_e = 0$.

On the even cycle C_L (edges indexed mod L, vertex v between edges $v-1$ and v) the kernel is spanned by $\sigma_e = (-1)^e$, and since $S_v = N_{v-1} + N_v$,

$$\frac{1}{S_v}\Big(\frac{1}{N_{v-1}} + \frac{1}{N_v}\Big) = \frac{1}{N_{v-1} N_v}, \qquad \langle M \rangle_t = \int_0^t \sum_v \frac{ds}{N_{v-1}(s) N_v(s)}.$$

3. Growth of every edge

Lemma 2. *Let d be the maximum degree and $\lambda < 2/(d+1)$. Almost surely $\liminf_{t\to\infty} N_e(t)/t^\lambda > 0$ for every edge e. In particular every count tends to infinity and $\langle M_\sigma\rangle_\infty < \infty$.*

Proof. Let $F_w(t)$ be the number of rings of w by time t. A ring of w adds to S_v only if $w = v$ or $w \sim v$, and then by one, so $S_v(t) \le d + \sum_{w\in\{v\}\cup N(v)} F_w(t)$. Since $F_w(t)/t \to 1$ a.s., for $\varepsilon > 0$ there is an a.s. finite T with $S_v(t) \le (d+1)(1+\varepsilon)t$ for all v and $t \ge T$. For $e = (u, v)$ the jump rate of N_e is $N_e(S_u^{-1} + S_v^{-1}) \ge \lambda' N_e/t$ on $[T, \infty)$, with $\lambda' = 2/((d+1)(1+\varepsilon))$. In the time $\tau = \log t$ this is a pure birth process with rate at least $\lambda' n$ at height n, so by thinning it dominates a Yule process of rate λ' started at time $\log T$ from $N_e(T) \ge 1$, and $e^{-\lambda'\tau} Y(\tau)$ has a positive a.s. limit. Taking ε small gives the first claim. For the second, every ring of v counts in S_v, so $S_v(t) \ge F_v(t)$, and the rate in Theorem 1 is at most $d\,\|\sigma\|_\infty^2/(F_v(t)\min_{e\ni v} N_e(t)) = O(t^{-1-\lambda})$. □

4. Convergence on the even cycle

Let $L \ge 4$ be even. In the time $\tau = \log t$ the normalised counts follow, up to a martingale error of order $t^{-1/2}$, the field

$$\frac{dx_e}{d\tau} = x_e\,\partial_e\mathcal{L}(x), \qquad \mathcal{L}(x) = \sum_v \log S_v(x) - \sum_e x_e, \tag{2}$$

where $\partial_e\mathcal{L} = S_u^{-1} + S_v^{-1} - 1$ for $e = (u, v)$, and $\frac{d}{d\tau}\mathcal{L} = \sum_e x_e(\partial_e\mathcal{L})^2 \ge 0$ [4]. Also $\frac{d}{d\tau}\sum_e x_e = L - \sum_e x_e$.

Lemma 3. *The equilibria of (2) in $\{\sum_e x_e = L,\ S_v > 0\ \forall v\}$ are*

1. *the segment $\Sigma = \{x^{(u)} : u \in [-1, 1]\}$, $x_e^{(u)} = 1 + (-1)^e u$, on which $S_v \equiv 2$;*
2. *finitely many points, each with an edge e such that $x_e = 0$ and $\partial_e\mathcal{L}(x) \ge 1/(2L)$.*

Proof. Let x be an equilibrium. A vertex with both edges at zero has $S_v = 0$, which is excluded. If every edge is positive, $1/S_u + 1/S_v = 1$ along every edge of the cycle, so S takes two values s, s' alternately; summing, $\sum_v S_v = 2\sum_e x_e = 2L$ gives $s + s' = 4$, hence $s = s' = 2$ and $x \in \Sigma$. Otherwise the positive edges form paths $w_0 w_1 \cdots w_k$ separated by single zero edges, and along such a path $S(w_i)$ alternates between s and $s' = s/(s-1)$. Since the edges outside the path at w_0 and w_k are zero, $x_1 = S(w_0) = s$, and solving along the path gives $x_{2i} = id$, $x_{2i+1} = s - id$ with $d = s' - s$. If k is odd the end condition $x_k = S(w_k) = s'$ forces $d = 0$, hence $k = 1$ and $x_1 = 2$. If k is even it forces $d = 2s/k$, and $1/s + 1/s' = 1$ gives $s = (2k+2)/(k+2)$; all x_j are then positive. So every block is either a single edge of weight 2 with $S = 2$ at both ends, or an even block of length k with $S = (2k+2)/(k+2)$ at both ends, and the configuration is determined by the block lengths. If all

blocks have length one, x alternates $2, 0$ and lies in Σ ($u = \pm 1$). Otherwise some even block of length $k \le L-1$ meets a zero edge e whose other end has $S \in \{2, (2k'+2)/(k'+2)\}$, so $\partial_e \mathcal{L} \ge \frac{k+2}{2k+2} + \frac{1}{2} - 1 = \frac{1}{2k+2} \ge \frac{1}{2L}$, the smallest value occurring when the other end has $S = 2$. □

Theorem 4. *On C_L, L even, almost surely $X(t) \to x^{(u_\infty)}$ with $|u_\infty| < 1$ and*

$$L \operatorname{artanh}(u_\infty) = M_\infty := \lim_{t\to\infty} \sum_e (-1)^e H(N_e(t) - 1).$$

Proof. By Lemma 2, $S_v(t) \ge F_v(t)$, so $X(t)$ eventually stays in the compact set $K = \{\sum x_e \le 2L,\ S_v \ge 1/2\}$, on which (2) is smooth. The jumps of X are of size $1/t$ and the martingale part has quadratic variation $O(1/t)$ per unit of $\log t$, so $X(e^\tau)$ is an asymptotic pseudotrajectory of (2) [7], and its limit set is a connected, internally chain transitive set. Since $\mathcal{L}$ is a strict Lyapunov function and by Lemma 3 takes finitely many values on the equilibria, the limit set lies in a connected set of equilibria, that is, either in Σ or at one point of type (2).

Type (2): suppose $X(t) \to x^*$ with $x^*_e = 0$ and $\partial_e \mathcal{L}(x^*) \ge 1/(2L)$. For $t \ge T$ the jump rate of N_e is at least $(1+\delta)N_e/t$ with $\delta = 1/(4L)$. The comparison of Lemma 2, run from the stopping time at which X leaves a small neighbourhood of x^* after T, shows that on the event of never leaving, $N_e(t) \ge c\, t^{1+\delta/2}$, so $X_e \to \infty$, a contradiction. Hence this event has probability zero for each T and each neighbourhood, and X does not converge to a type (2) point.

So the limit set lies in Σ. By (1) and Lemma 2, $\sum_e (-1)^e \log X_e(t) \to M_\infty$, finite a.s. At $x^{(u)}$ the left side equals $\frac{L}{2} \log \frac{1+u}{1-u} = L \operatorname{artanh}(u)$, which is strictly increasing in u and infinite at $u = \pm 1$. Every limit point therefore equals $x^{(u_\infty)}$ with $L \operatorname{artanh}(u_\infty) = M_\infty$. □

Remark 5. The same argument applies whenever the interior equilibria of a fibre $\{\sum_e \sigma_e \log x_e = m : \sigma \in \ker B\}$ are unique and boundary equilibria are transversally unstable. Uniqueness within a fibre is Birch's theorem for the transportation polytope $\{x > 0 : Bx = b\}$: the map $x \mapsto$ (projection of $\log x$ on $\ker B$) is a bijection from its interior onto $\ker B$.

5. The Gaussian limit on large even cycles

Write $Q_v = \int_0^\infty \frac{dt}{N_{v-1}(t) N_v(t)}$, so that $\langle M \rangle_\infty = \sum_v Q_v$ on C_L, and $Q_v^{\le T}$, $Q_v^{>T}$ for the integrals over $[0, T]$ and (T, ∞).

Lemma 6. *There is $\varepsilon(T) \to 0$ such that $\mathbb{E}\, Q_v^{>T} \le \varepsilon(T)$ for every even $L \ge 6$ and every v.*

Proof. Since $\max(N_{v-1}, N_v) \ge S_v/2$ and $S_v \ge F_v \vee 2$, $\frac{1}{N_{v-1}N_v} \le \frac{2}{F_v \vee 1}\big(\frac{1}{N_{v-1}} + \frac{1}{N_v}\big)$. Build the process from rate-one Poisson clocks at the vertices and independent uniform marks U attached to the rings; at a ring of w the left edge $w-1$ is chosen when $U < N_{w-1}/S_w$ and the right edge w when

$U \geq 1 - N_w/S_w$. For the edge $e = v$ let $G_w(s) = 2 + F_{w-1}(s) + F_w(s) + F_{w+1}(s) \geq S_w(s)$ and let $\tilde{N}_e$ start at 1 and increase at a ring of v when $U \geq 1 - \tilde{N}_e/G_v(s)$, and at a ring of $v+1$ when $U < \tilde{N}_e/G_{v+1}(s)$. If $\tilde{N}_e \leq N_e$ before a ring, every increase of $\tilde{N}_e$ is also an increase of N_e, so $N_e \geq \tilde{N}_e$ at all times. $\tilde{N}_e$ and F_v are functions of the clocks of $v-1, \dots, v+2$ and the marks at $v, v+1$, so the law of $\int_T^\infty \frac{2}{F_v \vee 1}(\tilde{N}_{v-1}^{-1} + \tilde{N}_v^{-1})\, dt$ does not depend on L, and it suffices to show its mean is finite. By Cauchy–Schwarz, $\mathbb{E}(F_v(t) \vee 1)^{-2} \leq Ct^{-2}$ and $\tilde{N}_e^{-2} \leq \tilde{N}_e^{-1}$, it is enough that $\mathbb{E}[\tilde{N}_e(t)^{-1}] \leq Ct^{-a}$ for some $a > 0$.

Let $s = \sqrt{t}$ and $\rho = \inf\{r \geq s : \max(G_v(r), G_{v+1}(r)) > 4r\}$, a stopping time for the clocks with $\mathbb{P}(\rho < \infty) \leq Ce^{-c\sqrt{t}}$ by Poisson large deviations. Couple $\tilde{N}_e$ on $[s, \rho)$ with a pure birth process $\hat{Y}$, $\hat{Y}(s) = 1$, increasing at a ring of w at time r with the same mark rule and probability $\min(1, \hat{Y}/(4r))$; then $\hat{Y} \leq \tilde{N}_e$ on $[s, \rho)$. Until $\hat{Y}$ exceeds $4r$ it is, in the time $\log r$, a Yule process of rate $1/2$, and once it does, $\hat{Y}(t) \geq 4s$. A Yule process of rate $1/2$ run for time $\frac{1}{2}\log t$ from 1 is geometric with parameter $p = t^{-1/4}$, with $\mathbb{E}[Y^{-1}] = p\log(1/p)/(1-p)$. Hence $\mathbb{E}[\tilde{N}_e(t)^{-1}] \leq Ce^{-c\sqrt{t}} + Ct^{-1/4}\log t + (4\sqrt{t})^{-1}$. □

Lemma 7. *For each T there is $q(T)$ with $\frac{1}{L}\sum_v Q_v^{\leq T} \to q(T)$ in L^2 as $L \to \infty$, and $q(T) \uparrow c < \infty$.*

Proof. Use the same clocks and marks on C_L and on $\mathbb{Z}$. The edge counts at v up to time T are determined by the rings and marks in the backward cluster of (v, T): the set of vertices reachable by a chain of rings at strictly decreasing times, each at a neighbour of the previous one. The radius of this cluster exceeds R only if a chain of R rings with decreasing times occurs within $[0, T]$, which has probability at most $\delta_R = 2(2T)^R/R!$. Let $q(T)$ be the mean of $Q_0^{\leq T}$ for the process on $\mathbb{Z}$. Since $0 \leq Q_v^{\leq T} \leq T$, $|\mathbb{E}Q_v^{\leq T} - q(T)| \leq T\delta_R$, and $Q_v^{\leq T}, Q_w^{\leq T}$ are independent up to probability $2\delta_R$ when $|v - w| > 2R + 2$. So $\mathrm{Var}(\frac{1}{L}\sum_v Q_v^{\leq T}) \leq T^2((4R+5)/L + 4\delta_R)$; let $L \to \infty$, then $R \to \infty$. Finally $q(T)$ is increasing in T, and $q(T) \leq 1 + \varepsilon(1)$, since $Q_0^{\leq 1} \leq 1$ and the proof of Lemma 6 applies verbatim on $\mathbb{Z}$. □

Theorem 8. *Let $c = \mathbb{E}\int_0^\infty \frac{dt}{N_{-1}(t)N_0(t)}$ for the critical WARM on $\mathbb{Z}$ with unit initial counts. As $L \to \infty$ through even values, $\sqrt{L}\,\mathrm{artanh}(u_\infty) \Rightarrow \mathcal{N}(0, c)$. In particular $u_\infty \to 0$ in probability at rate $L^{-1/2}$.*

Proof. By Lemmas 6 and 7, $\langle M\rangle_\infty/L \to c$ in L^1. The jumps of $M/\sqrt{L}$ are at most $L^{-1/2}$. The martingale central limit theorem for triangular arrays [6] gives $M_\infty/\sqrt{L} \Rightarrow \mathcal{N}(0, c)$, and Theorem 4 gives $\sqrt{L}\,\mathrm{artanh}(u_\infty) = M_\infty/\sqrt{L}$. □

Theorem 4 also gives the exact finite-L representation $u_\infty = \tanh(M_\infty/L)$ with $\mathbb{E}M_\infty = 0$ and $\mathbb{E}M_\infty^2 = \mathbb{E}\langle M\rangle_\infty$. Numerically $c \approx 1.64$ (Section 6).

6. Numerical checks and the infinite line

All simulations run the process exactly (rate-one vertex clocks, unit initial counts).

Even cycles. For R independent runs to time T we record $M(T)$, $\langle M\rangle_T$ and $\operatorname{artanh}(u)$, where u is read from $\sum_e (-1)^e \log X_e(T) = L \operatorname{artanh}(u)$.

L	T	R	mean M (s.e.)	$\operatorname{Var} M$	$\mathbb{E}\langle M\rangle_T$	$\mathbb{E}\operatorname{artanh}(u)^2 \cdot L$
100	10^3	400	0.30 (0.64)	161.6	164.3	1.61
400	10^3	200	−2.86 (1.76)	620	657	1.56
100	10^4	200	–	154.9	164.7	–

The quadratic variation per vertex is 1.64 at both sizes, so $c \approx 1.64$, and the difference $\sum_e (-1)^e \log X_e(T) - M(T)$ in (1) has root mean square 0.007 at $T = 10^4$, $L = 100$. At $T = 10^3$ the counts on C_{100} are not yet a pure alternation: the slow modes of (2) near wavenumber π decay like $t^{-\pi^2/L^2}$.

Small graphs. On K_4, where $\dim\ker B = 2$, the standard deviation of each X_e over 2000 runs stays near 0.35 between $T = 10^3$ and $T = 10^5$, while $\max_v |S_v/t - 2|$ falls from 0.25 to 0.08: the limit is a random point of a two-dimensional set of equilibria. On C_4 the limit is random on the segment Σ; the variance of $(X_0 + X_2)/4$ is 0.0564, which matches a $\mathrm{Beta}(b, b)$ law only for $b \approx 1.72$, so the limit law is not of the Pólya form. On the triangle ($\ker B = 0$) the spread decreases, more slowly than $t^{-1/4}$, towards $(1, 1, 1)$.

The line. On $\mathbb{Z}$ the kernel is replaced by a local conservation law. With $Y_e = (-1)^e H(N_e - 1)$ and $J_v = (-1)^v / S_v$, the computation in Theorem 1 gives

$$dY_e = -(J_{e+1} - J_e)\, dt + dM_e,$$

with M_e a martingale. Linearising (2) at $x \equiv 1$ gives the decay rate $(1 + \cos k)/2$ for the mode e^{ike}, which vanishes at $k = \pi$. Writing $x_e = 1 + (-1)^e u(e) + w(e)$ with u slowly varying, the field w relaxes to $(u^2)''/8$ and, formally, u obeys

$$\partial_\tau u = \tfrac{1}{4}(1 - u^2)\, \partial_e^2 u, \qquad \text{equivalently} \qquad \partial_\tau v = \tfrac{1}{4}\, \partial_e^2 \tanh v, \quad v = \operatorname{artanh} u,$$

which conserves $\int v$, the continuum form of the charge. Started from the noise of the early stage, this heat flow in $\tau = \log t$ predicts that the local alternating amplitude decays like $(\log t)^{-1/4}$. On a cycle of length 1001 run to $T = 10^5$ the standard deviation of X_e falls from 0.57 at $t = 12$ to 0.39 at $t = 8.8 \times 10^4$, and $(\log t)^{1/4}$ times it stays between 0.67 and 0.72, rising slowly (a fitted exponent of -0.18 in $\log t$ over this range, which a shift of τ by about 3 or the saturation of tanh would account for); the correlation of neighbouring edges tends to -0.92. These runs are consistent with Kleptsyn's conjecture, with convergence at the logarithmic rate the charges predict.

Acknowledgements and funding

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research

and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25).

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [8] and working together inside the Brain framework [10], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [9], the Math Brain repository [8] and the Brain framework repository [10].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In November 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, found the charge martingales of Theorem 1, proved the results of Sections 3–5, wrote and ran the simulations of Section 6, and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code is available at
`https://github.com/yspennstate/graph-urn-conserved-charges`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [8].

References

1. G. Aletti, A. Collevecchio, I. Crimaldi, M. Deijfen, S. Dereich, A. Ghiglietti, R. van der Hofstad, M. Holmes, D. Kious, V. Kleptsyn, B. Lodewijks, P. Mörters and B. Schapira, Open problems in stochastic reinforcement models, in 2023 MATRIX Annals, MATRIX Book Series, Springer, 2025, 477–498.
2. M. Benaïm, I. Benjamini, J. Chen, Y. Lima, A generalized Pólya's urn with graph based interactions, Random Structures Algorithms 46 (2015), 614–634.
3. J. Chen, C. Lucas, A generalized Pólya's urn with graph based interactions: convergence at linearity, Electron. Commun. Probab. 19 (2014).
4. R. van der Hofstad, M. Holmes, A. Kuznetsov, W. Ruszel, Strongly reinforced Pólya urns with graph-based competition, Ann. Appl. Probab. 26 (2016).

5. Y. Lima, Graph-based Pólya's urn: completion of the linear case, Stoch. Dyn. 16 (2016).
6. P. Hall, C. C. Heyde, Martingale Limit Theory and Its Application, Academic Press (1980).
7. M. Benaïm, Dynamics of stochastic approximation algorithms, Séminaire de Probabilités XXXIII, Lecture Notes in Math. 1709, Springer (1999).
8. Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, `https://github.com/yspennstate/math-brain-public`.
9. Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, `https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24`.
10. Y. Shmalo, *Brain Framework*, public software repository, 2026, `https://github.com/yspennstate/brain-framework-public`.